ZHAO Yuxi

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1 EDUCATION

Université Paris-Saclay **09/2024-10/2026**

M1 - Master in Electronics, Electrical Energy, and Automation (EEEA), 14.6/20, top1%

M2 - Automatic, Signal&Image Processing(ATSI)

Statistical Machine Learning and Reinforcement Learning, Nonlinear System Control, MIMO, Convex Optimization, Multi-Agent Systems

Université Toulouse III - Paul Sabatier **08/2020-07/2024**

Bachelor in Electronics, Electrical Energy, and Automation, 14.364/20, top10%

Industrial Information, Digital and Analog Electronics, Electric Machines, Principles of Communication, Linear Automation Systems, Signal Processing

Northeastern University(211&985), Liaoning, China(Double Degree) **09/2020-06/2024**

Bachelor in Materials Science and Engineering, 3.3/5.0, top15%
Physical Chemistry, Mechanical Properties of Materials, Fluid Mechanics, Linear Algebra, Dynamics, Mechanical Design

2 EXPERIENCE

Research Intern, R&D Institute — Anhui Mojia Zhichuang Robot Technology Co. (MOGA Humanoid Robotics), Wuhu, China **04/2026–08/2026**

Reinforcement learning and actuator system identification for a 30-DoF humanoid robot

- Identified actuator dynamics (armature, direction and velocity dependent friction/damping, communication delay) for MOGA's 7 custom motor types using the PACE framework, driving a CMA-ES optimization loop in Isaac Lab against chirp-sweep excitation data collected through a custom ROS 2 pipeline.
- Designed and implemented a multi-joint chirp-sweep excitation system in ROS 2 Control to support simultaneous independent chirp trajectories across arbitrary joint sets, each with its own frequency range, amplitude, and PD gains, while keeping non-excited joints locked at stand pose with high-stiffness PD control.
- Extended the actuator model to reduce average on-robot tracking RMSE compared to non-identified defaults, and showed that combined multi-joint identification outperforms isolated single-joint identification for load-bearing joints.
- Trained a PPO-based standing-balance policy (InstinctLab/Isaac Lab, 4096 parallel environments) that rejects upper-body disturbance motion, using domain randomization ranges centered on the identified actuator parameters rather than uncharacterized defaults, directly narrowing the sim-to-real gap for deployment.
- Tools: ROS 2, Isaac Lab, MuJoCo, PyTorch

Identification of nonlinear dynamic systems **12/2025-03/2026**

Study bioprocesses and model them as black-box or gray-box models, particularly using machine learning approaches. The focus is on the Physics-Informed Neural Network (PINN) approach.

Force Feedback System Design and Dynamics Modeling Project **04/2025-06/2025**

Multiphysics dynamics modeling, Multi-mode closed-loop control design, Modular System Validation

Robotics Project – Control using AED andASL systems **02/2024-06/2024**

Designed a control system for a mobile robot using Discrete Event Systems (AED) and Linear Systems Control (ASL). Developed a state machine and PID controllers based on system dynamics.

Trapped Charges in Polymers – Experimental Study **10/2023-12/2023**

Studied the energetic distribution oftrapped electrical charges in PEN thin-film polymers using LIMM and PSD techniques.

3 LANGUAGES & IT SKILLS

French : B1, English : B2, Chinese: Native

Programming & Tools : Python, C, Matlab, VHDL, Catia, ROS 2, Pytorch/TensorFlow, MuJoCo

4 PERSONAL PROFILE

I hold a dual degree in Electronics and Materials Science through a Chinese-French cooperative program. I am rigorous, autonomous, and enjoy solving engineering problems through theory and practice. Studying in both China and France has broadened my vision and strengthened my intercultural communication skills. I am passionate about learning, hands-on work, and innovation.